

Adelic Coefficients on the Full Split-Support Base Prime Transport, Localized Supports, and Synchronization

Programme derivations with the original human sources cited throughout

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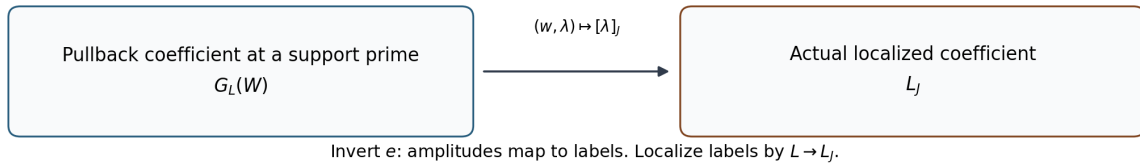
This reader constructs the entire divisor-indexed family of the original Connes–Consani adelic coefficient sheaves and its exact extension to an arbitrary bounded distributive support lattice. It retains the rational prime maps, original Haar measures, mixed-prime cohomology, actual support localizations, and both original-zeta endpoint contributions. The incoming finite heat/translation construction is applied to those same sheaves. The different projectors remain visible at support primes even though their additive group reflections agree. No identification of the resulting positive quotient metric with Weil’s form is assumed. The complete arguments are CDV1–25, ULC1–18, and CLS1–13. The appendices retain the support-localization and Mellin proofs used by them. The source archive also retains the complete predecessor sources and the unchanged original author archives.

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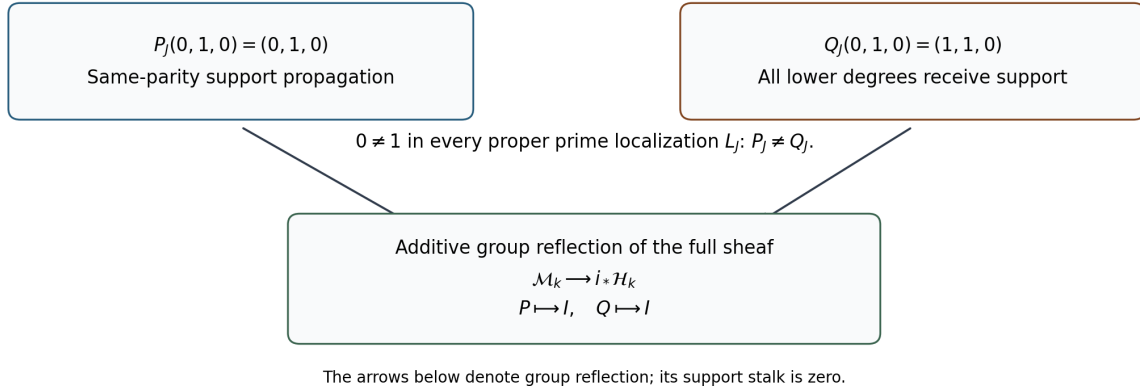
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The actual localized support retains the difference

$$Y = \operatorname{Spec} G_L(\mathbb{Z}), \quad S = \operatorname{Spec}_{\text{lat}} L, \quad J \in S$$



Independent degrees 0, 1, 2; test input $(0, 1, 0) \in L_J^3$



Proofs: ULC4-10; CLS3-8. Exact algebraic diagram, not a numerical sample.
 Adelic source: Alain Connes and Caterina Consani, arXiv:2501.06560v1, Section 4.

Figure 1: The exact source and support comparisons. The vector $(0, 1, 0)$ is a lattice-valued coefficient vector, not a vector of complex amplitudes. The bottom arrows refer to the amplitude reflection of the sheaf; at a support point its target is zero. ULC4-10 and CLS3-8 prove all displayed maps.

1 All rational-prime translates of the original adelic coefficient sheaf

Human source, original object, and scope. Alain Connes and Caterina Consani, *Knots, primes and class field theory*, arXiv:2501.06560v1, Section 4, original author file `arxiv-pisa.tex`, Lemma `localv`, Proposition `structure`, and Theorem `structure1`, construct the semilocal coefficient sheaf using the literal transition vectors $1_{\mathbb{Z}_p}$. Their author source, SHA-256 249fed347469cfaf6da97915d547c2ca704a4ba21cc339613713e570f14d5cea, has been read in full in this task. The proof below starts with those functions and keeps every rational scaling and Haar factor. It extends the programme calculation CCF8–34, whose complete source is retained, by making the nonunit-prime defect into a family with invertible maps. The original scalar function remains $\zeta(s)$; its complete local integrals are calculated at the end. No identification with a Weil positive form is assumed.

1.1 The actual family and its sheaf maps

Put $X = \operatorname{Spec} \mathbb{Z}$ and $V = \mathcal{S}(\mathbb{A}_{\mathbb{Q}})$. Here this Bruhat–Schwartz space is the algebraic restricted tensor product of the real Schwartz space and the finite locally constant compactly supported function spaces, with standard vectors $1_{\mathbb{Z}_p}$. Thus an element is a finite sum of tensors with only finitely many nonstandard finite factors. Let

$$\mathfrak{D} = \bigoplus_p \mathbb{Z}, \quad k = (k_p)_p, \quad c_{p,k_p} = 1_{p^{k_p} \mathbb{Z}_p}, \quad N(k) = \prod_p p^{k_p} \in \mathbb{Q}_{>0}. \quad (\text{CDV1})$$

Every product indexed by the support of k is finite. Write $V^{(p)}$ for the restricted tensor product at every place except p , including the real factor, and set

$$V_{p,k} = \mathbb{C} c_{p,k_p} \otimes V^{(p)} \subset V. \quad (\text{CDV2})$$

For $U = X \setminus F$, F a finite set of finite primes, define

$$\mathcal{O}_k(U) = \bigcap_{p \notin F} V_{p,k} \subset V, \quad \mathcal{O}_k(\emptyset) = 0. \quad (\text{CDV3})$$

All restriction maps for nonempty inclusions are inclusions of these actual subspaces. Explicitly its sections are the tensors with arbitrary finite factors at $p \in F$ and the specified c_{p,k_p} at every other finite prime, with arbitrary real Schwartz factor and finite linear sums.

Here is a proof of this explicit intersection and of the sheaf property. For a finite collection of primes define commuting idempotents on V by

$$P_{p,k} f(x_p, x^{(p)}) = p^{k_p} c_{p,k_p}(x_p) \int_{p^{k_p} \mathbb{Z}_p} f(y, x^{(p)}) dy, \quad \operatorname{vol}(\mathbb{Z}_p) = 1. \quad (\text{CDV4})$$

The volume of this ball is p^{-k_p} , so $P_{p,k}^2 = P_{p,k}$ and its image is exactly $V_{p,k}$. Different p act on different tensor coordinates and therefore commute. For a given f , choose a finite set containing its nonstandard factors and $\operatorname{supp} k$. Outside that set the conditions are automatic; inside it, being fixed by every required projection is equivalent to belonging to the image of their product. This gives precisely the factors in CDV3. Every nonempty open in X contains the generic point, so compatible sections on a covering are the same element $f \in V$. Their local conditions say exactly that $f \in V_{p,k}$ for every point of the covered open. This is CDV3, proving existence and uniqueness of gluing. The generic stalk is V , and the stalk at p is $V_{p,k}$ because every finite set away from p can be removed in its neighbourhoods. At $k = 0$ this is the original Connes–Consani sheaf with its unchanged transition factors.

The sheaf is an algebra under the original pointwise multiplication: $c_{p,k_p}^2 = c_{p,k_p}$, so products preserve every required local factor. This statement does not assert a multiplicative structure on the quotient vector spaces introduced below.

For $r \in \mathbb{Q}^\times$ write $d(r) = (v_p(r))_p$ and keep the actual dilation

$$(\alpha_r f)(x) = f(r^{-1}x), \quad \alpha_r : \mathcal{O}_k \longrightarrow \mathcal{O}_{k+d(r)}. \quad (\text{CDV5})$$

At p , multiplication by the unit part of r fixes every ball, so $\alpha_r c_{p,k_p} = c_{p,k_p+v_p(r)}$. At all other coordinates the same formula acts on the actual Schwartz factors. This proves CDV5 on every open and its compatibility with restrictions. Its inverse is $\alpha_{r^{-1}}$, and $\alpha_r \alpha_q = \alpha_{rq}$. These are algebra isomorphisms between the specified sheaves. The stabilizer of the index k consists exactly of $r = \pm 1$, by unique factorization of a rational number. The family indexed by $\mathfrak{D}$ is retained; no member is replaced by another without this explicit map.

For crossed-product coefficients retain

$$B = V \rtimes \mathbb{Q}^\times, \quad B_{p,k} = \bigoplus_{u: v_p(u)=0} V_{p,k} U_u, \quad (fU_u)(gU_v) = f\alpha_u(g)U_{uv}. \quad (\text{CDV6})$$

The sheaf $\mathcal{A}_k(X \setminus F)$ is the intersection of the $B_{p,k}$ for $p \notin F$. The argument proving CDV3 works coefficient by coefficient because every crossed-product element has a unique finite sum in the group labels. It yields exactly the coefficient sheaf $\mathcal{O}_k$ crossed with the units of $\mathbb{Z}[F^{-1}]$. The unit condition makes the pointwise coefficient product preserve the prescribed local balls. Since rational dilations commute,

$$\alpha_r^B \left(\sum_u f_u U_u \right) = \sum_u \alpha_r(f_u) U_u, \quad \alpha_r^B : \mathcal{A}_k \xrightarrow{\sim} \mathcal{A}_{k+d(r)} \quad (\text{CDV7})$$

preserves the displayed multiplication, all original group labels, and every restriction. Its inverse is $\alpha_{r^{-1}}^B$.

1.2 Every mixed-prime cohomology component

For either $W = V$ or $W = B$, use $W_{p,k} = V_{p,k}$ or $B_{p,k}$. Define

$$D_k(W) = \bigoplus_p W/W_{p,k}, \quad \delta_k w = ([w]_{p,k})_p, \quad C_k(W) = [W \xrightarrow{\delta_k} D_k(W)]. \quad (\text{CDV8})$$

The arrow lands in the direct sum: only finitely many primes occur in k , the coefficient tensors, or the nonunit valuations of a group label. The sheaf sequence

$$0 \longrightarrow \mathcal{H}_k \longrightarrow c_X W \longrightarrow \bigoplus_p i_{p*}(W/W_{p,k}) \longrightarrow 0 \quad (\text{CDV9})$$

is exact, where $\mathcal{H}_k = \mathcal{O}_k$ or $\mathcal{A}_k$. At p it is the quotient sequence and at the generic point its last term is zero; its kernel on an open is precisely the defining intersection. The constant sheaf is flasque because all nonempty opens are connected and its restriction maps are identity maps. The last sheaf has finite-support sections on every nonempty open: its generic germ is zero, so a section vanishes on a cofinite neighbourhood of the generic point, leaving only finitely many closed points. Its restrictions forget components and are onto. Thus it is also flasque, and CDV9 computes

$$H^0(X, \mathcal{H}_k) = \ker \delta_k, \quad H^1(X, \mathcal{H}_k) = D_k(W)/\delta_k W, \quad H^j(X, \mathcal{H}_k) = 0 \quad (j \geq 2). \quad (\text{CDV10})$$

This uses the definition of sheaf cohomology through an acyclic resolution; flasque sheaves are acyclic for global sections. All maps and terms of that resolution have been specified.

The complete mixed-place decomposition can be written without altering the original Haar lengths of the local vectors. Put

$$E_{p,k} = \left\{ f \in \mathcal{S}(\mathbb{Q}_p) : \int_{p^{k_p} \mathbb{Z}_p} f = 0 \right\}, \quad \mathcal{S}(\mathbb{Q}_p) = \mathbb{C}_{c_{p,k_p}} \oplus E_{p,k}. \quad (\text{CDV11})$$

The two projections are exactly CDV4 and $I - P_{p,k}$. Expanding a finite tensor gives an algebraic direct sum

$$V = \bigoplus_{T \text{ finite}} V_{T,k}, \quad V_{T,k} = \mathcal{S}(\mathbb{R}) \otimes \bigotimes_{p \in T} E_{p,k} \otimes \bigotimes'_{p \notin T} \mathbb{C}_{c_{p,k_p}}. \quad (\text{CDV12})$$

Only finitely many terms occur for any input; projection by the corresponding finite product of $P_{p,k}$ and $I - P_{p,k}$ proves uniqueness. A T component survives in $V/V_{p,k}$ precisely when $p \in T$. For a crossed-product term $V_{T,k}U_u$ it survives precisely at $T_u = T \cup \{p : v_p(u) \neq 0\}$. Therefore the actual differentials in CDV8 are the diagonal maps on these components, and

$$\begin{aligned} H^1(X, \mathcal{O}_k) &= \bigoplus_{|T| \geq 2} V_{T,k} \otimes (\mathbb{C}^T / \Delta \mathbb{C}), \\ H^1(X, \mathcal{A}_k) &= \bigoplus_{u, T: |T_u| \geq 2} V_{T,k} U_u \otimes (\mathbb{C}^{T_u} / \Delta \mathbb{C}). \end{aligned} \quad (\text{CDV13})$$

Indeed an empty marked set has zero target, a one-element set has surjective diagonal, and for every larger finite set the cokernel is exactly its marked coordinates modulo the diagonal. This proves CDV13 by actual component maps, rather than dimension counting.

Every rational dilation induces a chain isomorphism

$$C_k(W) \xrightarrow{\alpha_r} C_{k+d(r)}(W), \quad [w]_{p,k} \mapsto [\alpha_r w]_{p,k+d(r)}, \quad (\text{CDV14})$$

with group labels unchanged in the crossed product. Well-definedness follows from CDV5 and CDV7, and commutation with δ follows componentwise. The inverse is the corresponding map for r^{-1} . The local projections conjugate exactly: $\alpha_r P_{p,k} \alpha_{r^{-1}} = P_{p,k+d(r)}$. This follows either from their ranges and orthogonality, or directly from the change of variables in CDV4. Hence every mixed component T , every marked prime, and every diagonal relation is transported by CDV14. No cohomology component is dropped in this prime action.

1.3 The original positive metric and the nonunit defect

The additive adelic Haar measure is the original product of dx at infinity and measures with $\text{vol}(\mathbb{Z}_p) = 1$. In particular

$$\|c_{p,k_p}\|^2 = p^{-k_p}, \quad \|\alpha_r f\|^2 = |r|_\infty \prod_p p^{-v_p(r)} \|f\|^2 = \|f\|^2. \quad (\text{CDV15})$$

The product formula here follows immediately by writing $r = \pm \prod p^{v_p(r)}$. Every factor in the preceding product is retained. The quotient $V/V_{p,k}$ is isometrically represented by $(I - P_{p,k})V$, since CDV4 is the original Haar orthogonal projection. Equip $D_k(V)$ with the sum of these squared quotient norms. In $D_k(B)$, at a coefficient with $v_p(u) = 0$ use this projection, and at $v_p(u) \neq 0$ use its full original Haar norm; sum over the finite coefficient and prime labels. CDV14 is an isometry for these metrics, by CDV15 and the conjugation identity.

Completion and quotient do not delete the mixed arithmetic part. The orthogonal expansion CDV12, retaining all factors p^{-k_p} , gives

$$\overline{D_k(V)} / \overline{\delta_k V} = \widehat{\bigoplus_{|T| \geq 2} V_{T,k} \otimes (\Delta \mathbb{C})^\perp}. \quad (\text{CDV16})$$

To verify the closures, finite tensors are dense in each Hilbert summand by its definition. Each finite collection of diagonal components is the image of their sum in V . Conversely the bounded orthogonal projection on every marked-coordinate factor annihilates $\delta_k V$, so its closure has no component outside the diagonal. These two inclusions prove the displayed equality. On $\mathbb{C}^T$ the diagonal projection is exactly $(c_p)_p \mapsto (|T|^{-1} \sum_p c_p)_p$; its denominator remains. The same proof with T_u gives CDV16 for B with every U_u retained. The original algebraic quotient injects into the Hilbert quotient, since a finite vector has zero orthogonal projection exactly when its components were already diagonal. These are particular positive quotient metrics, not an identification with the original Weil trace.

If instead one insists on the fixed target $V/V_{p,k}$, the nonunit action has a nonzero defect. For an integer m and $f = c_{p,k_p} \otimes g$, its representative is

$$(I - P_{p,k})\alpha_{p^m} f = \left(c_{p,k_p+m} - p^{-\max(0,m)} c_{p,k_p} \right) \otimes \alpha_{p^m}^{(p)} g. \quad (\text{CDV17})$$

Indeed the overlap of the two balls has volume $p^{-\max(k_p, k_p+m)}$ and CDV4 supplies the multiplier p^{k_p} . Taking its original squared norm gives

$$\|(I - P_{p,k})\alpha_{p^m} f\|^2 = p^{-k_p} (1 - p^{-|m|}) \|g\|^2. \quad (\text{CDV18})$$

For $m \geq 0$ the finite local squared norm is $p^{-k_p}(p^{-m} - p^{-2m})$; for $m < 0$ it is $p^{-k_p}(p^{-m} - 1)$. The other places contribute p^m by the same product formula, yielding CDV18 in both cases. It vanishes exactly at $m = 0$ for nonzero g . Thus the obstruction to a fixed-space endomorphism is a calculated vector; CDV14 constructs the invertible action on its actual target family.

1.4 Fourier maps with every local scalar retained

Use the conductor- $\mathbb{Z}_p$ additive character at p and $e^{-2\pi i xy}$ at infinity. Character orthogonality proves

$$\widehat{c_{p,k_p}} = p^{-k_p} c_{p,-k_p}. \quad (\text{CDV19})$$

The annihilator of $p^{k_p} \mathbb{Z}_p$ is $p^{-k_p} \mathbb{Z}_p$: on it the integral is the ball volume; off it, translating by a point on which the character is nontrivial forces the integral to be zero. Tensoring gives a vector-sheaf isomorphism $\mathcal{F} : \mathcal{O}_k \rightarrow \mathcal{O}_{-k}$. Its inverse is Fourier followed by $x \mapsto -x$; this follows from Fourier inversion on the real Schwartz factor and character orthogonality on finite local quotients. In semilocal coordinates the transition includes the full scalar $\prod_{p \notin F} p^{-k_p}$; it is not the bare map replacing each standard vector by $c_{p,-k_p}$.

On the original crossed-product multiplication the target is convolution, and the exact isomorphism is

$$\sum_u f_u U_u \mapsto \sum_u \widehat{f}_u U_{u^{-1}}, \quad \mathcal{F} \alpha_r = \alpha_{r^{-1}} \mathcal{F}, \quad \mathcal{F} \vartheta_\lambda = \lambda \vartheta_{\lambda^{-1}} \mathcal{F}. \quad (\text{CDV20})$$

Here ϑ_λ acts only on the real coordinate by $x_\infty \mapsto \lambda^{-1} x_\infty$, $\lambda > 0$. The last two equalities follow by change of variables in the Fourier integrals; the rational product of moduli is 1, the real modulus is λ . Fourier of a pointwise product is convolution by Fubini on these functions. Therefore the image of $f \alpha_u(g) U_{uv}$ is $\widehat{f} * \alpha_{u^{-1}}(\widehat{g}) U_{u^{-1}v^{-1}}$, which is exactly the target product. It preserves the nonunit valuation conditions and takes k to $-k$. For example the image vector $p^{-k_p} c_{p,-k_p}$ is a convolution idempotent: its convolution square is $p^{-2k_p} p^{k_p} c_{p,-k_p}$, equal to itself. This verifies why the retained scalar matters to the algebra map. All these arrows induce the specified chain and cohomology maps because they send each local coefficient subspace to its target.

1.5 Original zeta and the two endpoint functionals

For $a > 0$ define the actual tensor

$$f_{k,a}(x) = e^{-\pi(x_\infty/a)^2} \prod_p c_{p,k_p}(x_p). \quad (\text{CDV21})$$

The two additive moments, computed in the original measures, are

$$f_{k,a}(0) = 1, \quad \int_{\mathbb{A}} f_{k,a}(x) dx = a \prod_p p^{-k_p} = a/N(k). \quad (\text{CDV22})$$

The real Gaussian integral is a by substitution; the finite integrals are exactly the ball volumes. Thus these are separate endpoint data, not both replaced by one. Rational dilation gives $\alpha_r f_{k,a} = f_{k+d(r),|r|a}$, and real scaling gives $\vartheta_\lambda f_{k,a} = f_{k,\lambda a}$. CDV22 then proves that the rational map preserves both moments whereas the real map multiplies only the integral moment by λ .

For each finite prime use multiplicative Haar measure with $\text{vol}^\times(\mathbb{Z}_p^\times) = 1$, and at infinity use $dx/|x|$. For $\text{Re } s > 1$, define the adelic multiplicative integral as the absolutely convergent product of its local integrals. Their complete evaluation is

$$\begin{aligned} \int_{\mathbb{Q}_p^\times} c_{p,k_p}(x) |x|_p^s d^\times x &= \sum_{j=k_p}^{\infty} p^{-js} = \frac{p^{-k_p s}}{1 - p^{-s}}, \\ \int_{\mathbb{R}^\times} e^{-\pi(x/a)^2} |x|^s \frac{dx}{|x|} &= a^s \pi^{-s/2} \Gamma(s/2), \\ Z(f_{k,a}, s) &= a^s N(k)^{-s} \pi^{-s/2} \Gamma(s/2) \zeta(s). \end{aligned} \quad (\text{CDV23})$$

For the first line partition the support into annuli $p^j \mathbb{Z}_p^\times$; multiplicative invariance makes each annulus have measure 1. The geometric series converges for $\text{Re } s > 0$. For the second integrate over both signs and substitute $y = \pi(x/a)^2$; the factor of two and the Jacobian give exactly the displayed Gamma factor. At the global step $\sum_p p^{-\text{Re } s} < \infty$ bounds the product. Unique prime factorization identifies its absolutely convergent expansion with $\sum_{n \geq 1} n^{-s} = \zeta(s)$. Hence the original function is recovered, with every factor shown, by

$$\zeta(s) = a^{-s} N(k)^s \frac{\pi^{s/2}}{\Gamma(s/2)} Z(f_{k,a}, s). \quad (\text{CDV24})$$

No endpoint polynomial $s(s-1)$ was inserted into the arithmetic function. Meromorphic continuation here uses the original-zeta formula ZTC10 with its separate $1/(s-1)$ and $-1/s$ terms. For the actual rational map in CDV22, both a and $N(k)$ are multiplied by $|r|$, so their two factors in CDV23 cancel exactly. Changing only the finite-place ball would leave p^{-ms} and would be a different arithmetic operation. This is the full prime and archimedean comparison behind that cancellation.

1.6 Full lattice coefficients and their actual limits

Let L be any nontrivial bounded distributive lattice, and retain $G_L(W) = \{(0, \lambda) : \lambda \in L\} \cup (W \times \{1_L\})$. Every linear map above has the exact lift $(w, \lambda) \mapsto (Tw, \lambda)$. Addition uses label join and scalar multiplication label meet; substitution proves compatibility and shows that zeros of amplitudes do not delete their labels. Every inverse already displayed lifts by the same formula. In particular the quotient congruence is exactly the following $G_L(\mathbb{C})$ -semimodule congruence; it is not a quotient-algebra assertion, since $W_{p,k}$ need not be an ideal:

$$(w, \lambda) \sim (w', \lambda') \iff \lambda = \lambda' \text{ and } w - w' \in W_{p,k}, \quad G_L(W)/\sim \cong G_L(W/W_{p,k}). \quad (\text{CDV25})$$

Its classes are the fibres of the surjective quotient map; the relation is compatible with both operations since $W_{p,k}$ is a vector subspace. Direct-sum formulas lift with one common label, not independently chosen labels on the target factors. For sheaves the sectionwise construction is sheafified; stalks are $G_L(\mathcal{H}_{k,x})$. To verify that description, filtered colimits commute with the carrier construction: a nonzero amplitude is represented with top label; a labelled zero is represented by that same labelled zero at any stage, and equality is exactly equality at a later

stage together with equality of labels. Sheafification leaves stalks unchanged. All maps and inverse identities therefore survive sheafification.

This coefficient lift does not remove the supported-zero prime. For $G_L(\mathbb{Z})$ its element $e = (0, 1_L)$ generates the ideal of all labelled zeros, since $e(a, \lambda) = (0, \lambda)$. The amplitude map $q : G_L(\mathbb{Z}) \rightarrow \mathbb{Z}$ sends this ideal to (0) , and $q^{-1}(0) = (e)$ is prime: if $q(x)q(y) = 0$ in the domain $\mathbb{Z}$, one amplitude is zero. This proves the actual prime statement, retaining every zero label. The arithmetic prime (p) similarly has inverse image $q^{-1}(p)$ containing (e) strictly. The rest of the support spectrum and its stalks remain part of the base; no scalar Mellin evaluation is asserted to recover them. CDV25 specifies the coefficient maps that can be applied on that base, while its full support cohomology requires the sheaf extension constructed separately.

2 The Connes–Consani coefficient family on the actual arbitrary support base

Sources and scope. The human adelic source is Alain Connes and Caterina Consani, *Knots, primes and class field theory*, Section 4, author-source labels `localv`, `structure`, and `structure1`. CDV1–25 supplies the complete divisor-indexed extension and proofs, retaining the original local Haar measures and real factor. FSC1–9 and FSC33–34 supply the support topology and localization maps; their full proofs accompany this chapter. Their foundation is the pinned Split Support Geometry v11 source, Definition `def:lattice-split` and the subsequent ideal and prime classification. This chapter derives the actual coefficient modules and the arbitrary-support cohomology comparison. It does not identify an idempotent semimodule with an abelian sheaf.

2.1 The base and its two types of open

Let L be a nontrivial bounded distributive lattice, possibly infinite, and retain

$$X = \operatorname{Spec} \mathbb{Z}, \quad S = \operatorname{Spec}_{\text{lat}} L, \quad Y = \operatorname{Spec} G_L(\mathbb{Z}) = S \sqcup X. \quad (\text{ULC1})$$

The points of S are proper prime lattice ideals J , represented in the semiring spectrum by $\{(0, \lambda) : \lambda \in J\}$. The points of X are $q^{-1}(\mathfrak{p})$, where $q(a, \lambda) = a$. An ideal containing a top-supported element contains $e = (0, 1_L)$: add that element to its product with $(-1, 1_L)$. Multiplication by e then supplies all labelled zeros. Its amplitudes form a ring ideal. Every ideal without a top-supported element is a proper lattice ideal of zero labels. Primality is respectively ring or meet primality. This proves the stated partition of the points. Its basic opens are $D((n, 1_L)) = S \cup D_X(n)$ for $n \neq 0$ and $D((0, \lambda)) = D_L(\lambda) \subset S$. Consequently every open of Y is either an open $V \subset S$ or

$$W_U = S \cup U \quad (\emptyset \neq U \subset X \text{ open}). \quad (\text{ULC2})$$

All nonempty opens of X contain the generic point η and are cofinite. Write $i : X \hookrightarrow Y$ and $j : S \hookrightarrow Y$. The map $r : Y \rightarrow X$ is the identity on X and sends every point of S to η . It is continuous since $r^{-1}U = W_U$ for nonempty U , and $ri = \text{id}$. Nontriviality of L implies $S \neq \emptyset$ by prime-ideal separation, whose proof is included in FSC6.

The actual structure sheaf has

$$\mathcal{O}_Y(W_U) = G_L(\mathcal{O}_X(U)), \quad \mathcal{O}_Y|_S = \mathcal{O}_S, \quad \mathcal{O}_S(D_L(\lambda)) = \downarrow \lambda, \quad (\mathcal{O}_S)_J = L_J. \quad (\text{ULC3})$$

Here L_J is the localization with equality $[\lambda]_J = [\mu]_J$ precisely when $\lambda \wedge \nu = \mu \wedge \nu$ for some $\nu \notin J$. On basic support opens the restriction is $a \mapsto a \wedge \lambda$. Finite covers glue by joins and distributivity; compactness of the basic opens reduces an arbitrary covering to this assertion. On W_U , the generic arithmetic germ fixes one rational amplitude and one label, and the amplitude must be regular at every included prime; this proves the first formula. Restriction to S retains the label and localizes it. These observations prove ULC3 with the original structure sheaf, not a constant substitute.

2.2 The actual localized coefficient semimodule

Let $\mathcal{H}_k$ be either CDV's $\mathcal{O}_k$ or $\mathcal{A}_k$ on X , with generic complex vector space $W = V$ or B and stalks $W_{p,k}$. It is an $\mathcal{O}_X$ -module by multiplication by the rational scalar in each regular section. This scalar action preserves each complex subspace and every restriction.

Define a sheaf $\mathcal{M}_k$ of $\mathcal{O}_Y$ -semimodules by

$$\mathcal{M}_k(W_U) = G_L(\mathcal{H}_k(U)), \quad \mathcal{M}_k(V) = \mathcal{O}_S(V) \quad (V \subset S). \quad (\text{ULC4})$$

On arithmetic opens, restrictions restrict the amplitude and keep the label. Restriction from W_U to a support open sends (w, λ) to the section represented by λ , with all its localizations retained.

On support opens use the actual restrictions of $\mathcal{O}_S$. The scalar operation on an arithmetic open is $(a, \mu)(w, \lambda) = (aw, \mu \wedge \lambda)$ and addition uses join. On support opens scalar operation is meet and addition is join. All these formulas commute with restrictions.

Here is a direct gluing proof. A cover of W_U has some arithmetic members W_{U_a} covering U . Any two have nonempty arithmetic intersection, so compatible sections have one common label and their amplitudes glue in $\mathcal{H}_k$. A nonzero glued amplitude has nonzero restriction somewhere, hence top label. The arithmetic member already covers all of S ; compatibility forces each purely support member to carry precisely the restriction of that common label. Thus the glued section is unique and belongs to ULC4. Covers of a support open glue in $\mathcal{O}_S$. This proves the sheaf assertion and the scalar operations on the original carrier. Its stalks are

$$(\mathcal{M}_k)_\eta = G_L(W), \quad (\mathcal{M}_k)_p = G_L(W_{p,k}), \quad (\mathcal{M}_k)_J = L_J. \quad (\text{ULC5})$$

The first two follow by the filtered neighbourhood colimits and CDV's stalk descriptions. At J the neighbourhoods contained in S are cofinal and give the last formula.

This is the actual localization of the coefficient carrier at support. For every $w \in W$, multiplication by e sends (w, λ) to $(0, \lambda)$. Inverting the idempotent e makes it the unit, so

$$G_L(W)[e^{-1}] \simeq L, \quad G_L(W)_{P_J} \simeq L_J. \quad (\text{ULC6})$$

To verify the first isomorphism rather than only its surjectivity, the support map has the section $\lambda \mapsto (0, \lambda)$; every localized element equals its product with e , and equality of these zero labels is unchanged by multiplication by e . Further inversion of the zero labels outside J gives exactly the relation in ULC3. Every top-supported nonzero integer already maps to the unit of that localized lattice. No additional relation arises from those integers. This proves the second isomorphism.

Retain separately the inverse-image coefficient sheaf

$$\mathcal{P}_k = r^{-1}G_L(\mathcal{H}_k), \quad \mathcal{P}_k \longrightarrow \mathcal{M}_k. \quad (\text{ULC7})$$

Here $G_L(\mathcal{H}_k)$ on X has the section formula used in ULC4; its sheaf proof follows from the same common-generic-point argument. The arrow is the identity on arithmetic opens. On a support open it sends a locally constant section of $G_L(W)$ to its support label and then to its localizations in $\mathcal{O}_S$. It commutes with restrictions because a section on W_U restricts in $\mathcal{P}_k$ to its constant generic germ, whereas in $\mathcal{M}_k$ it restricts to its localized label. Its stalk map is the identity at arithmetic points and

$$G_L(W) \longrightarrow L \longrightarrow L_J, \quad (w, \lambda) \longmapsto \lambda \longmapsto [\lambda]_J \quad (\text{ULC8})$$

at a support prime. It is surjective on stalks. The precise fibre relation at J is existence of $\nu \notin J$ with $\lambda \wedge \nu = \mu \wedge \nu$; amplitudes impose no further condition. This comparison is linear over the corresponding sheaf map $r^{-1}G_L(\mathcal{O}_X) \rightarrow \mathcal{O}_Y$. One cannot make the unchanged $\mathcal{P}_k$ into an $\mathcal{O}_Y$ -semimodule extending that coefficient action at a support point: e is a unit there but kills each nonzero amplitude under the original action. The map ULC8 computes the required localization rather than assuming it away.

2.3 Retaining the support ideal and the exact amplitude reflection

Let $\mathcal{D} = j_*\mathcal{O}_S$. Its sections on W_U are L and its sections on $V \subset S$ are $\mathcal{O}_S(V)$. There are actual semimodule maps

$$\mathcal{D} \xrightarrow{z} \mathcal{M}_k \xrightarrow{\sigma} \mathcal{D}, \quad \sigma z = \text{id}, \quad q : \mathcal{M}_k \longrightarrow i_*\mathcal{H}_k. \quad (\text{ULC9})$$

On W_U , $z(\lambda) = (0, \lambda)$, $\sigma(w, \lambda) = \lambda$, and $q(w, \lambda) = w$. On support opens the first two maps are the identity, and the last map has zero target. On $\mathcal{D}$ the scalar action is induced by $\mathcal{O}_Y \rightarrow \mathcal{D}$; on the amplitude target it is induced by $\mathcal{O}_Y \rightarrow i_*\mathcal{O}_X$. These formulas prove compatibility with both operations and every restriction. They give

$$e\mathcal{M}_k = z\mathcal{D} = q^{-1}(0), \quad \mathcal{M}_k / \sim_{z\mathcal{D}} \simeq i_*\mathcal{H}_k, \quad (\text{ULC10})$$

where $x \sim_{z\mathcal{D}} y$ means locally $x + z(u) = y + z(v)$ for support sections u, v . Indeed multiplication by e is precisely $z\sigma$. Two sections $(w, \lambda), (w', \mu)$ on W_U with equal amplitude obey $(w, \lambda) + z(\mu) = (w', \mu) + z(\lambda)$. Conversely adding zero-amplitude sections cannot change the amplitude. On a support open the entire module is $z\mathcal{D}$. The map q is onto on sections, using $(w, 1_L)$ as a lift. This proves the quotient description, including its original congruence.

The sheaf associated with additive group completion of $\mathcal{M}_k$ is exactly $i_*\mathcal{H}_k$. Every additive map to a sheaf of abelian groups kills each idempotent zero label because $z + z = z$. It then identifies the two sections in the preceding displayed relation and factors uniquely through q . Conversely the target is already an abelian sheaf, and each w is in the image. This proves the universal property locally and hence for sheaves. It does not identify $\mathcal{M}_k$ itself with its group completion. The retract $z\mathcal{D}$, all its localized labels, and the map ULC8 remain explicit parts of the original object.

The image of (q, σ) inside $i_*\mathcal{H}_k \times \mathcal{D}$ has on W_U the exact constraint $w \neq 0 \Rightarrow \lambda = 1_L$ and has the whole support factor on support opens. It is injective on every stalk by ULC5. Thus an unrestricted direct product is not the original carrier.

2.4 Cohomology for an arbitrary support spectrum

For any abelian sheaf F on Y there is a natural isomorphism

$$r_*F \simeq i^{-1}F. \quad (\text{ULC11})$$

The map is induced by restriction from W_U to neighbourhoods of U along i . Its stalk at $x \in X$ is the identity on F_x : all neighbourhoods of x in Y are exactly the W_U for neighbourhoods U in X . This proves the isomorphism and shows that r_* is exact. It preserves injectives because its left adjoint r^{-1} is exact. Since $\Gamma(Y, F) = \Gamma(X, r_*F)$, an injective resolution gives

$$R\Gamma(Y, F) \simeq R\Gamma(X, i^{-1}F). \quad (\text{ULC12})$$

For $F = r^{-1}\mathcal{H}_k$, the right side is represented by the actual CDV complex

$$C_k = [W \xrightarrow{\delta_k} D_k], \quad D_k = \bigoplus_p W/W_{p,k}, \quad \delta_k w = ([w]_{p,k})_p. \quad (\text{ULC13})$$

For $F = i_*\mathcal{H}_k$, the same conclusion follows either from ULC11–12 or from the exactness of closed-embedding direct image. Consequently the amplitude reflection in ULC10 has $H^1 = D_k/\delta_k W$, including every mixed-prime summand of CDV13, and vanishes in degrees at least two. No finiteness of L has been used.

There is more information in the relative comparison for the *inverse-image vector sheaf* $r^{-1}\mathcal{H}_k$. Its restriction to S is the constant sheaf $c_S W$, because $r|_S$ factors through η . This is not the support restriction $\mathcal{O}_S$ of the original semimodule $\mathcal{M}_k$. For clarity define an explicit acyclic resolution of $c_S W$ as follows. For an abelian sheaf E put $T(E)(V) = \prod_{x \in V} E_x$ with projection restrictions. This is a sheaf and is flasque; the germ map $E \rightarrow T(E)$ is injective. Set $E_0 = c_S W$, $E_{n+1} = \text{coker}(E_n \rightarrow T(E_n))$, and use the composites $T(E_n) \rightarrow E_{n+1} \rightarrow T(E_{n+1})$. The resulting sheaf sequence is exact by its kernel and cokernel definitions, and all its terms are flasque. Let $B^n = \Gamma(S, T(E_n))$ with differential d_B . Then $B^\bullet$ computes $R\Gamma(S, c_S W)$. This construction retains every infinite product over the actual points of S .

The constant section gives $c : W \rightarrow \ker(d_B : B^0 \rightarrow B^1)$. The relative cohomology with support in the closed arithmetic stratum is computed by

$$W \xrightarrow{d^0} D_k \oplus B^0 \xrightarrow{d^1} B^1 \xrightarrow{-d_B} B^2 \xrightarrow{-d_B} \dots, \quad d^0 w = (\delta_k w, cw), \quad d^1(d, b) = -d_B b. \quad (\text{ULC14})$$

Here degrees start at zero. To justify the relative comparison, take an injective resolution on Y . Its restrictions to the open S are injective since extension by zero is an exact left adjoint. Each

injective sheaf is flasque, so restriction of its global sections to S is onto. The kernel complex therefore computes sections with support in X , and the short exact sequence of complexes yields the fibre of global restriction. Apply the CDV flasque resolution to $r^{-1}\mathcal{H}_k$: inverse image is exact, its constant first term is $c_Y W$, and its closed-prime term has zero restriction to S . These terms are acyclic for global sections by ULC12, whose direct images are precisely the two CDV flasque terms. Restriction of this resolution to S has just $c_S W$ in degree zero. Its comparison with the displayed resolution $B^\bullet$ is the constant germ map in degree zero and zero in degree one. Thus the fibre complex, with the sign convention shown, is exactly ULC14. Its differential squares to zero since $d_B c = 0$.

As S is nonempty, c is injective. Writing $\mathrm{LC}(S, W) = \Gamma(S, c_S W)$, calculation of the kernels and images in ULC14 gives

$$\begin{aligned} H_X^0(Y, r^{-1}\mathcal{H}_k) &= 0, \\ H_X^1(Y, r^{-1}\mathcal{H}_k) &= (D_k \oplus \mathrm{LC}(S, W)) / \{(\delta_k w, cw) : w \in W\}, \\ H_X^{j+1}(Y, r^{-1}\mathcal{H}_k) &= H^j(S, c_S W) \quad (j \geq 1). \end{aligned} \tag{ULC15}$$

In particular there is an exact sequence

$$0 \longrightarrow D_k \xrightarrow{d \mapsto [(d, 0)]} H_X^1 \longrightarrow \mathrm{LC}(S, W) / cW \longrightarrow 0. \tag{ULC16}$$

Injectivity of the first map follows from injectivity of c . The last map sends $[(d, b)]$ to $[b]$; its kernel is the displayed image, since subtracting $(\delta_k w, cw)$ removes any constant b . Choosing a point $s_0 \in S$ gives the explicit splitting $[(d, b)] \mapsto (d - \delta_k(b(s_0)), [b])$; its inverse uses the unique representative $b - b(s_0)$ of a locally constant class vanishing at s_0 . The splitting is not asserted to be independent of that point. The natural map to ordinary $H^1(Y, r^{-1}\mathcal{H}_k)$ sends $[(d, b)]$ to $[d] \in D_k / \delta_k W$.

No identity replacing $H^j(S, c_S W)$ by $W \otimes H^j(S, \mathbb{C})$ is used for an arbitrary infinite S . The full resolution above specifies its coefficient maps without commuting tensor products through infinite products.

2.5 Prime transport on all these objects

For every nonzero rational r , CDV5 and CDV7 induce the exact maps

$$\mathcal{M}_k \xrightarrow{\alpha_r} \mathcal{M}_{k+d(r)}, \quad (w, \lambda) \mapsto (\alpha_r w, \lambda) \text{ on } W_U, \quad \text{id on support opens.} \tag{ULC17}$$

They commute with restrictions, scalar multiplication, z, σ, q and ULC7. They have inverse $\alpha_{r^{-1}}$ and preserve every localization relation of L . On the arithmetic amplitude complexes they are precisely CDV14. On the constant-support vector resolution they apply α_r to each stalk coordinate; functoriality of the products and cokernels above defines every higher map. Thus ULC14–16 commute with prime transport, including their complete mixed-prime defects. The support action on the actual localized semimodule is the identity, while the auxiliary constant vector sheaf carries α_r there. ULC8 is their exact connecting map.

For the original tensors $f_{k,a}$, CDV22–24 now give on this same arithmetic stratum

$$\begin{aligned} f_{k,a}(0) &= 1, & \int f_{k,a} &= a/N(k), \\ Z(f_{k,a}, s) &= a^s N(k)^{-s} \pi^{-s/2} \Gamma(s/2) \zeta(s), & \Re s > 1, \\ \alpha_r f_{k,a} &= f_{k+d(r), |r|a}, & N(k+d(r)) &= |r|N(k). \end{aligned} \tag{ULC18}$$

Both endpoint moments and all Mellin factors are retained. At the support stalk ULC8 sends this top-labelled coefficient to $1 \in L_J$, independently of its amplitude. That local statement follows from the localization already proved; it is not a Mellin inversion theorem reconstructing L from

a scalar function. The earlier fixed-prime quotient defect remains CDV17–18 with its full norm; ULC17 supplies its invertible family target over the actual base.

For a three-element chain $0 < a < 1$, the prime $J = \{0\}$ inverts a , so the zero labels a and 1 become equal in L_J , whereas the other prime $\{0, a\}$ leaves all three labels distinct. For every $w \neq 0$, the pullback stalk retains $(w, 1)$, while the localized stalk at either prime has its image 1 . Globally ULC4 still distinguishes $(w, 1)$, $(0, 1) = e$ and $(0, 0) = \tau$. This explicit example records the localization, its fibres and its arithmetic comparison without reducing the entire support spectrum to a single point or a Boolean coordinate.

3 Heat and translation projectors on the localized adelic coefficient family

Provenance and original objects. The incoming programme source is *Dynamical Synchronization, Heat Remapping, and the Full Supported Cauchy–Weil Form*, 23 September 2026, RECONSTRUCTION.tex, ST1–12 and SY1–16. Its archive is retained unchanged with SHA-256 70810361fd8fd66f9cc7154ed919bc5c0c15d003e3ec72504b173bbb24aeed15. Those finite-source sections were read in full. Their identities are proved below in the exact receiving sheaves; neither their later analytic claims nor the sign of a Weil form is assumed. The adelic coefficient input is Alain Connes and Caterina Consani’s *Knots, primes and class field theory*, Section 4, extended with full proofs in CDV1–25 and ULC1–18. The original arithmetic function remains ζ , with the exact factors of ULC18 and ZTC10. Finite formal-coordinate heat here is not an assertion that those coefficients follow Newman heat.

3.1 Finite matrices on the original semimodule sheaf

Use the actual localized sheaf $\mathcal{M}_k$ of ULC4 on $Y = \text{Spec } G_L(\mathbb{Z})$ and fix $N \geq 2$. Its independent coefficient source is $\mathcal{M}_k^{N+1}$, ordered by $1, Z, \dots, Z^N$. A constant split complex scalar $(c, \lambda) \in G_L(\mathbb{C})$ acts on arithmetic-open sections by $(w, \mu) \mapsto (cw, \lambda \wedge \mu)$ and on a support stalk L_J by meeting with $[\lambda]_J$. These maps commute with restrictions by ULC4 and with the $\mathcal{O}_Y$ action. Nonzero rational coefficients, including their factorial denominators, therefore define the stated endomorphisms without treating them as regular rational functions at every arithmetic prime.

Let $D_{i,i+1} = (i+1)^\bullet$ and every structural zero be τ . For split parameters a, b define

$$F_{a,b} = \sum_{n=0}^N \frac{(-aD^2 + bD)^n}{n!}, \quad (F_{a,b})_{ij} = \sum_{q=0}^{\lfloor (j-i)/2 \rfloor} \left(\frac{(-1)^q j!}{i! q! (j-i-2q)!} \right)^\bullet a^q b^{j-i-2q} \quad (j \geq i). \quad (\text{CLS1})$$

Below the diagonal the entries are τ . In the first formula the reciprocal factorial is fully supported, and the zeroth power is the free identity matrix. Each generator strictly lowers degree, so all omitted higher terms are the global zero matrix. To obtain the second formula, select q factors of $-aD^2$ and $j-i-2q$ factors of bD , then use the binomial coefficient and $D^{j-i}(Z^j) = j!Z^i/i!$. This keeps every sign and denominator. Commuting the generators and using the finite binomial identity gives

$$F_{a,b}F_{c,d} = F_{a+c,b+d}, \quad F_{\tau,\tau} = I. \quad (\text{CLS2})$$

This identity is valid in the semiring of matrices: the binomial proof uses distributivity, not subtraction of supported zeros.

At $a = (0, \lambda)$, $b = (0, \mu)$, every positive-degree amplitude vanishes but its original label remains. An odd degree drop has label μ because every term contains b and the term $q = 0$ has exactly that label. A positive even drop has the endpoint labels λ and μ , with all intermediate labels their meet, so its resulting label is $\lambda \vee \mu$. Thus

$$P = F_{e,\tau}, \quad Q = F_{e,e} = F_{\tau,e}, \quad P^2 = P, \quad Q^2 = Q, \quad PQ = QP = Q. \quad (\text{CLS3})$$

At an arithmetic stalk, these maps keep each amplitude and change the independent labels by

$$(P\lambda)_i = \bigvee_{j \geq i, j-i \text{ even}} \lambda_j, \quad (Q\lambda)_i = \bigvee_{j \geq i} \lambda_j. \quad (\text{CLS4})$$

This follows directly from matrix multiplication: the off-diagonal entries are zero-amplitude elements, so only the diagonal term contributes an amplitude, while joins record all admissible labels. Their images have respectively nested same-parity labels and a single nested flag. They are sheaf retractions onto their images.

3.2 Localization preserves the projectors' difference

At a support prime J , ULC8 sends every nonzero rational scalar and e to $1 \in L_J$, while τ remains zero. Consequently the localized P_J and Q_J are exactly the join operators CLS4 on $(L_J)^{N+1}$, now with their localized labels. They are different at every support prime: the input with entry 1 only at degree one and zero elsewhere has output degree zero equal to 0 under P_J and 1 under Q_J . A proper prime localization has $0 \neq 1$: an equality would require an element outside J to meet 0 and 1 equally, forcing that element to be $0 \in J$. Thus the example is nonzero in every support stalk.

The additive group reflection instead sends both projectors to the identity on $(i_*\mathcal{H}_k)^{N+1}$: every off-diagonal entry in them has amplitude zero. It follows, by applying the explicit CDV complex coordinatewise, that

$$H^j(P) = H^j(Q) = \text{id} \quad \text{on } H^j(X, \mathcal{H}_k)^{N+1}, \quad j \geq 0, \quad (\text{CLS5})$$

where these cohomology maps are those of the amplitude reflection. Every mixed-prime summand of CDV13 is included. The equality is not a claim that $P = Q$ on $\mathcal{M}_k^{N+1}$; their explicit support-stalk discrepancy proves the contrary. This gives the actual location of the information lost by ordinary additive cohomology in this calculation.

For supported $t, b \in \mathbb{C}$, CLS2 gives

$$F_{t^\bullet, b^\bullet} F_{(-t)^\bullet, (-b)^\bullet} = Q. \quad (\text{CLS6})$$

Also $FQ = QF = F$, since adding e to a supported parameter leaves that parameter unchanged. Hence these maps form a group on $Q\mathcal{M}_k^{N+1}$ with identity Q . On a support stalk all these supported-parameter matrices have the same image Q_J : in CLS1 every positive degree has at least one top-labelled term, and localization retains its label even when its amplitude cancels. They therefore act as the identity on the localized projective image. On the full free localized source, their common action is the nonidentity projector Q_J . The arithmetic amplitude group is the usual invertible polynomial heat/translation action.

3.3 The common-label comparison and its exact fibres

Define $\mathcal{M}_k[Z]_{\leq N}^{\text{one}}$ by the construction ULC4 applied to the vector sheaf $\mathcal{H}_k[Z]_{\leq N}$. Its arithmetic-open sections have one common label for the entire polynomial; its support restriction is $\mathcal{O}_S$. There is a surjective sheaf map

$$\eta_N : \mathcal{M}_k^{N+1} \longrightarrow \mathcal{M}_k[Z]_{\leq N}^{\text{one}}, \quad ((w_i, \lambda_i)) \longmapsto \left(\sum_i w_i Z^i, \bigvee_i \lambda_i \right). \quad (\text{CLS7})$$

On support opens it is the join of the coordinate sections. A nonzero polynomial forces at least one nonzero coefficient, hence top label; so the image belongs to the original carrier. Distributivity proves compatibility with scalar meets, and joins prove additivity. A polynomial with top label lifts by assigning top label to each coefficient. A zero polynomial with any label lifts by placing that label at degree zero and τ elsewhere. This proves surjectivity and compatibility with support restrictions. Its exact fibre relation is equality of every amplitude coefficient and equality of the total join; at a support stalk it is equality of the join in L_J .

Diagonal entries in CLS1 retain each input label at its own degree, while other entries contribute labels below existing ones. Therefore the total join is unchanged. The amplitude calculation is the ordinary polynomial exponential, giving

$$\eta_N F_{a,b} = \left[e^{-\text{amp}(a)\partial_Z^2 + \text{amp}(b)\partial_Z} \right]^\uparrow \eta_N, \quad \eta_N P = \eta_N Q = \eta_N. \quad (\text{CLS8})$$

On the common-label target the displayed linear operator keeps its single label, and on support it is the identity. This is the exact map under which the different projectors become equal.

3.4 Synchronization on every divisor-indexed coefficient source

Fix $d \geq 2$ and the monomial basis of total degree at most N in $s_1, \dots, s_d$, with independent copies of $\mathcal{M}_k$. The cyclic map β replaces Z^j by $(s_1 + \dots + s_d)^j$, with the literal multinomial coefficients. All nonzero coefficients are fully supported and all structural zeros are τ . Since each $D_i D_j \beta = \beta \partial_Z^2$, the ordinary amplitude covariance C acts by

$$\exp \left(\frac{1}{4} \sum_{i,j} C_{ij} D_i D_j \right) \beta = \beta \exp \left(\frac{1}{4} (\mathbf{1}^\top C \mathbf{1}) \partial_Z^2 \right). \quad (\text{CLS9})$$

For the original three matrices retain all entries and factors:

$$C_{\text{ind}} = \frac{t}{d} I, \quad C_{\text{sync}} = \frac{t}{d^2} \mathbf{1} \mathbf{1}^\top, \quad C_{\text{rel}} = \frac{t}{d} \left(I - \frac{\mathbf{1} \mathbf{1}^\top}{d} \right). \quad (\text{CLS10})$$

Lift only the independent off-diagonal structural zeros as τ , and every entry of the other two matrices as supported, as in the incoming source. Their off-diagonal sums are e , not τ . For each of these specified lifted matrices $\tilde{C}$, define $\mathcal{U}_C^\dagger = \exp(\frac{1}{4} \sum_{i,j} \tilde{C}_{ij} D_i D_j)$ on the stated total-degree source. The finite exponential uses the same fully supported reciprocal factorials and free identity as CLS1. The exact identity on the coefficient sheaf is therefore

$$\mathcal{U}_{\text{sync}}^\dagger \mathcal{U}_{\text{rel}}^\dagger = \mathcal{U}_{\text{ind}}^\dagger P_\Sigma, \quad P_\Sigma = \exp \left(\frac{e}{4} (D_1 + \dots + D_d)^2 \right). \quad (\text{CLS11})$$

Indeed the sum of the lifted generators is the sparse independent generator plus the displayed supported-zero generator: on the diagonal adding e does not change an already supported entry, and off the diagonal it supplies the missing supported cancellation. The finite commuting exponential proof of CLS2 gives the product identity. The matrix P_Σ has the identity on its diagonal, e at every coordinatewise lowering of positive even total degree, and τ elsewhere. These are exactly all terms of powers of the displayed derivative; each admissible path has a nonzero rational coefficient and hence the specified label. Composing two such paths again gives precisely that relation, proving $P_\Sigma^2 = P_\Sigma$.

The cyclic formula retains the projector:

$$P_\Sigma \beta = \beta P, \quad \mathcal{U}_{\text{rel}}^\dagger \beta = \beta P. \quad (\text{CLS12})$$

For a source monomial Z^j , β has top-supported coefficients at every monomial of total degree j . For any lower same-parity monomial choose a degree- j monomial above it coordinatewise. It receives a supported contribution under P_Σ ; no other degree or parity can receive one. Its amplitude remains zero in that lower degree. This proves the first equality entry by entry. For the second, the relative amplitude generator annihilates β by $\mathbf{1}^\top C_{\text{rel}} \mathbf{1} = 0$, while its fully supported entries supply the same admissible degree-lowering labels. This proves the second equality with its original support, including $t = 0^\bullet$.

Localization of CLS11 retains a visible difference. On the input supported only at $s_1 s_2$, P_Σ contributes 1 to the constant label at every support stalk. Sparse independent heat has no such constant contribution, since no D_i^2 can act on that monomial. The synchronized and relative product does have it, as CLS11 states. The amplitude of this extra term is zero before localization, and the ordinary group reflection sends P_Σ to the identity. Thus it is present on the actual support stratum even though it is absent in the original scalar amplitude equation.

Finally the rational-prime maps α_r of ULC17 commute with every matrix here. Each matrix acts on formal degrees by complex scalars and labels; α_r is complex linear on coefficients and fixes the labels. Hence every matrix entry commutes, proving

$$\alpha_r^{N+1} F_{a,b} = F_{a,b} \alpha_r^{N+1}, \quad \alpha_r P_\Sigma = P_\Sigma \alpha_r, \quad \alpha_r \beta = \beta \alpha_r, \quad k \mapsto k + d(r). \quad (\text{CLS13})$$

These identities hold between the actual divisor-indexed targets, with the original prime and archimedean scaling in CDV22–24. They do not assert that finite polynomial differentiation descends through an unspecified spectral quotient or changes the zeros of the original zeta function. The receiving maps, the localized projectors and the full mixed-prime amplitude cohomology are now specified on the same coefficient family.

A A common sheaf for spherical, tropical, and full support coefficients

A.1 Sources and the exact base

The human sources are Alain Connes and Caterina Consani, *Geometry of the Arithmetic Site*, original `arithmetic-site_Adv_final1.tex`, specifically Definition `site`, Theorem `structure2`, Remark `caseno`, Theorem `thmspz`, and Proposition `sheavesonspz0`; and their *On the Absolute Geometry of Spec Z and the Fargues–Fontaine curve*, original `FF.tex`, Proposition `prop12`, the monoid-algebra adjunction in Section 2.3, and the Frobenius formula at source lines 150–152. The arithmetic-site source was read here at lines 598–704 and 900–1017, without a claim to exhaustive reading; FF was previously read completely, lines 1–1764. The spherical functor H is the explicit coordinate functor of Connes–Consani’s *Absolute algebra and Segal’s Γ -rings*, original `F1_corr.tex`, formula `functsum` and Lemma `sssalg`. That source was previously read completely, lines 1–1336. The coefficient foundation is *The Clankers, Split Support Geometry*, Version 11, Definition `def:lattice-split` and its complete ideal and prime classification, in the complete pinned original source. The constructions below give the common diagram for the companion calculations ACF and ASL, with the required definitions and proofs included here.

Let L be a nontrivial bounded distributive lattice. It need not be finite or Boolean. For a nonzero commutative ring R retain

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in R\}, \\ (a, \lambda) + (b, \mu) &= (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu), \\ z_\lambda &= (0, \lambda), \quad \tau = z_{0_L}, \quad e = z_{1_L}, \quad 1 = (1_R, 1_L), \quad q(a, \lambda) = a, \quad \sigma(a, \lambda) = \lambda. \end{aligned} \tag{FSC1}$$

Put $S = G_L(\mathbb{Z})$, $X = \operatorname{Spec} S$, $Y = \operatorname{Spec} \mathbb{Z}$, and $A = \operatorname{Spec}_{\text{lat}} L$. The additive semiring prime spectrum has points

$$P_J = \{z_\lambda : \lambda \in J\} \quad (J \in A), \quad Q_{\mathfrak{p}} = q^{-1}(\mathfrak{p}) \quad (\mathfrak{p} \in Y). \tag{FSC2}$$

For completeness, an ideal containing a top-supported element contains e : add its product by $(-1, 1_L)$ to itself if its amplitude is nonzero. Then $ez_\lambda = z_\lambda$ puts the whole zero fibre in it, and its amplitudes form a ring ideal. All other ideals are exactly the proper lattice ideals in the zero fibre. For an amplitude ideal primality is ring primality; for a support ideal it is meet primality. This proves FSC2 directly. Their basic opens are

$$D_S(z_\lambda) = D_L(\lambda), \quad D_S((n, 1_L)) = A \cup i(D_{\mathbb{Z}}(n)), \quad A = D_S(e), \quad i(Y) = V_S(e). \tag{FSC3}$$

Here $i(\mathfrak{p}) = Q_{\mathfrak{p}}$ is the closed inclusion and $j : A \hookrightarrow X$ the open inclusion. Every open meeting $i(Y)$ is uniquely $W_U = A \cup i(U)$ for a nonempty open $U \subseteq Y$; the other opens are exactly the opens $V \subseteq A$. This follows from FSC3 because every nonempty open of Y contains its generic point and all support points are generizations of every arithmetic point. There is a continuous retraction

$$r(P_J) = r(Q_{(0)}) = \eta, \quad r(Q_{(p)}) = (p), \quad ri = \operatorname{id}. \tag{FSC4}$$

Indeed $r^{-1}D_{\mathbb{Z}}(n) = D_S((n, 1_L))$ for $n \neq 0$; $r^{-1}\emptyset = \emptyset$. These constitute the basis test. In particular $r^{-1}U = W_U$ for every nonempty open U .

A.2 The support sheaf, its direct image, and all their sections

For a lattice prime J define

$$L_J = L[(L \setminus J)^{-1}], \quad [\lambda]_J = [\mu]_J \iff \exists \nu \notin J : \lambda \wedge \nu = \mu \wedge \nu. \tag{FSC5}$$

Every invertible idempotent is the unit; hence this formula follows from the localization equality, and every fraction is represented by its numerator. It makes explicit the possible loss of a label in an individual support stalk.

Lemma A.1 (Sections of the support sheaf). *The restriction $\mathcal{O}_A = j^{-1}\mathcal{O}_X$ of the actual localization sheaf of S satisfies*

$$\mathcal{O}_A(D_L(\lambda)) = \downarrow \lambda, \quad a|_{D_L(\mu)} = a \wedge \mu \quad (\mu \leq \lambda), \quad (\mathcal{O}_A)_J = L_J. \quad (\text{FSC6})$$

In particular $\Gamma(A, \mathcal{O}_A) = L$. For every open $V \subseteq A$ its sections are exactly

$$\{(a_\lambda)_{D(\lambda) \subseteq V} : a_\lambda \leq \lambda, \quad a_\lambda \wedge \mu = a_\mu \text{ whenever } \mu \leq \lambda \text{ and } D(\lambda) \subseteq V\}. \quad (\text{FSC7})$$

The two operations are coordinatewise join and meet.

Proof. First $S[e^{-1}] = L$ because $e^2 = e$ makes $e = 1$ and $e(a, \lambda) = z_\lambda$. Then $L[\lambda^{-1}] = \downarrow \lambda$ by $a \mapsto a \wedge \lambda$, with unit λ . Indeed the equality of localized elements is exactly equality after meeting with λ . To verify the sheaf assertion, the prime-ideal separation theorem for bounded distributive lattices gives an injective map $\lambda \mapsto D_L(\lambda)$ preserving finite joins and meets, and these basic opens are compact. Here is the needed argument. If an ideal and filter are disjoint, an ideal maximal subject to avoiding the filter exists by Zorn's lemma and is prime: if it contains $a \wedge b$ but contains neither a nor b , adjoining either meets the filter, and distributivity makes the meet of those two filter elements belong to the original ideal, a contradiction. Apply this with the ideal $\downarrow \mu$ and filter $\uparrow \lambda$ when $\lambda \not\leq \mu$ to obtain separation. For compactness, a cover $D(\lambda) \subseteq \bigcup_i D(\mu_i)$ without a finite subcover would leave the ideal generated by all μ_i disjoint from $\uparrow \lambda$. The same prime would contradict the cover. For a finite cover $\lambda = \bigvee_i \lambda_i$, sections $a_i \leq \lambda_i$ agree on overlaps precisely when $a_i \wedge \lambda_j = a_j \wedge \lambda_i$. Their unique gluing is $\bigvee_i a_i$, by distributivity. Compactness reduces any cover of a basic open to this case. Thus FSC6 defines the localization sheaf on a basis and has the stalk FSC5. Restricting a section to every basis open gives FSC7; conversely those compatible values glue and are unique. This proves the formula for all opens, including $V = A$. $\square$

Set $\mathcal{D} = j_*\mathcal{O}_A$. There is an actual unital restriction map $d : \mathcal{O}_X \rightarrow \mathcal{D}$. Its sections and stalks are

$\mathcal{D}$	$W_U = A \cup i(U)$	$V \subseteq A$	$\mathcal{D}_{Q_p} = L, \quad \mathcal{D}_{P_J} = L_J.$	(FSC8)
$\mathcal{O}_X$	L	$\mathcal{O}_A(V)$		
	$G_L(\mathcal{O}_Y(U))$	$\mathcal{O}_A(V)$		

On W_U , d is σ ; on V , it is the identity. To justify the remaining entry for $\mathcal{O}_X$, localizing at an arithmetic prime gives $G_L(\mathbb{Z}_p)$, including $G_L(\mathbb{Q})$ at $Q_{(0)}$: all denominators there are nonzero integer amplitudes with top support, so labels never change. A section over W_U has one rational amplitude and one label at $Q_{(0)}$. Generization from each $Q_{(p)}$ injects $G_L(\mathbb{Z}_{(p)}) \hookrightarrow G_L(\mathbb{Q})$, forcing its amplitude to lie in $\mathcal{O}_Y(U)$ and its label to be the same. At all support points the section is determined by the germ at $Q_{(0)}$, and equals the localization of that label. Conversely the corresponding element of $G_L(\mathcal{O}_Y(U))$ gives these local fractions. This proves FSC8. Every neighbourhood of an arithmetic point contains all of A , so the direct-image stalk there is L ; at a support point it is the ordinary support stalk. Restriction from W_U to V sends (a, λ) to the section represented by λ , whose component on $D(\mu) \subseteq V$ is $\lambda \wedge \mu$.

Write $\underline{L}_X$ for the constant sheaf: its sections are the locally constant maps into the discrete set L , with pointwise lattice operations. There is a canonical unital sheaf map

$$\ell : \underline{L}_X \longrightarrow \mathcal{D}, \quad \ell_{Q_p} = \text{id}_L, \quad \ell_{P_J}(\lambda) = [\lambda]_J. \quad (\text{FSC9})$$

On W_U a locally constant map is constant: every point of A is a generization of $Q_{(0)}$ and every other arithmetic point specializes from $Q_{(0)}$, whereas a locally constant map into a discrete space is constant along either relation. Thus its sections there are L . On $V \subseteq A$ send a locally constant label function to the section whose germ at J is the localized label there; it is locally represented by a single label and therefore is a section of $\mathcal{O}_A$. This proves FSC9 on sections and under restrictions.

B Original arithmetic theta sums and both Mellin endpoints

The original functions here are $F_t(s) = \sum_{n \geq 0} (n+1+t)^{-s}$ for real $t > -1$ and $\Re s > 1$, and $F_0 = \zeta$. This is the complete ZTC2–10 calculation from the retained original-zeta reconstruction. The Gaussian Poisson identity used at ZTC9 is proved directly below. Its classical kernel provenance is Brad Rodgers and Terence Tao, arXiv:1801.05914v5, source equations `phidef`, `htdef`, `hoz`, `sas`. No heat deformation of zeta is needed for this Mellin comparison.

For real $t > -1$ and $x > 0$, retain the actual shifted frequencies and put

$$K_t(x) = \sum_{n=0}^{\infty} e^{-\pi(n+1+t)^2 x}. \quad (\text{ZTC2})$$

This is a scalar analytic representation of the stated arithmetic family. No assertion that it determines the full support lattice is part of this definition. For fixed $a = 1+t > 0$, monotonicity of the summand as a function of n gives

$$K_t(x) \leq e^{-\pi a^2 x} + \int_0^{\infty} e^{-\pi(a+y)^2 x} dy \leq 1 + \frac{1}{2\sqrt{x}}. \quad (\text{ZTC3})$$

For $x \geq 1$ the same bound after retaining the first exponential gives $K_t(x) \leq C_a e^{-\pi a^2 x}$, since $(a+n)^2 - a^2 \geq n^2$ and $C_a = \sum_{n \geq 0} e^{-\pi((a+n)^2 - a^2)} < \infty$. Consequently the following integral converges absolutely for $\Re s > 1$:

$$\int_0^{\infty} x^{s/2-1} K_t(x) dx = \pi^{-s/2} \Gamma(s/2) F_t(s), \quad F_t(s) = \frac{\pi^{s/2}}{\Gamma(s/2)} \int_0^{\infty} x^{s/2-1} K_t(x) dx. \quad (\text{ZTC4})$$

Indeed, with $\sigma = \Re s > 1$, the sum of the absolute termwise integrals equals $\pi^{-\sigma/2} \Gamma(\sigma/2) \sum_{n \geq 0} (n+a)^{-\sigma}$. Fubini therefore applies. Substitution $y = \pi(n+a)^2 x$ in each term proves the displayed identity with its full Gamma and pi factors. Continuation outside that domain means the meromorphic continuation proved in HTB4, not an assertion that the unsubtracted integral converges there.

There is also an exact inverse recovering the kernel, not only the arithmetic function. For every real $c > 1$,

$$K_t(x) = \frac{1}{4\pi i} \int_{c-i\infty}^{c+i\infty} \pi^{-s/2} \Gamma(s/2) F_t(s) x^{-s/2} ds. \quad (\text{ZTC4a})$$

Here the integral is absolutely convergent. To prove the inversion with its constant, set $\sigma = c/2 > 0$ and $g(v) = \exp(\sigma v - e^v)$. This function and all its derivatives are integrable on $\mathbb{R}$: at $-\infty$ they are bounded by a constant times $e^{\sigma v}$, and at $+\infty$ by a polynomial in e^v times $g(v)$. The substitution $u = e^v$ identifies its Fourier transform with $\Gamma(\sigma - iy)$. Repeated integration by parts therefore bounds that Gamma factor by $C_N(1+|y|)^{-N}$ for every fixed N . Fourier inversion gives $e^{-u} = (2\pi i)^{-1} \int_{\sigma-i\infty}^{\sigma+i\infty} \Gamma(w) u^{-w} dw$. One may verify this inversion directly by first multiplying the Fourier integrand by $e^{-\epsilon y^2}$: the resulting expression is the convolution of g with the Gaussian approximate identity, which converges to g ; the stated integrable transform bound permits dominated convergence as $\epsilon \downarrow 0$. Apply the identity to $u = \pi(n+1+t)^2 x$, then put $s = 2w$. The resulting factor is $1/(4\pi i)$. Finally sum in n . Interchange is justified by $\sum (n+1+t)^{-c} < \infty$ and the integrable Gamma bound. This proves ZTC4a. ZTC4 and ZTC4a are mutual recovery formulas for the specified original arithmetic family and its scalar theta sums; they do not recover a support lattice from a scalar function.

Differentiation in t is locally uniform for t in compact subsets of $(-1, \infty)$ and x in compact subsets of $(0, \infty)$, by Gaussian majorants for a polynomial times e^{-cn^2} . It gives the exact formula

$$\partial_t K_t(x) = -2\pi x \sum_{n=0}^{\infty} (n+1+t) e^{-\pi(n+1+t)^2 x}. \quad (\text{ZTC5})$$

For $\operatorname{Re} s > 0$, integration of its absolute value is bounded by a constant times $\sum (n+a)^{-\sigma-1}$, locally uniformly in t . More precisely its Mellin transform is

$$\int_0^\infty x^{s/2-1} \partial_t K_t(x) dx = -2\pi^{-s/2} \Gamma(s/2+1) F_t(s+1) = -s\pi^{-s/2} \Gamma(s/2) F_t(s+1). \quad (\text{ZTC6})$$

Thus the original arithmetic shift generator is represented directly, without substituting the heat generator for it. The stated equality extends meromorphically by HTB4–5.

The separated sheet and its entire missing summand are visible before any transform:

$$K_1(x) = K_0(x) - e^{-\pi x}, \quad D_t(s) = \zeta(s) - F_t(s), \quad D_1(s) = 1, \quad (\text{ZTC7})$$

$$\pi^{-s/2} \Gamma(s/2) D_t(s) = \int_0^\infty x^{s/2-1} (K_0(x) - K_t(x)) dx \quad (\operatorname{Re} s > 1). \quad (\text{ZTC8})$$

ZTC7 follows by deleting the $n = 1$ summand of K_0 ; ZTC8 follows by subtracting two absolutely convergent integrals. At $t = 1$ the integral is exactly $\pi^{-s/2} \Gamma(s/2)$, so its inverse returns the actual constant 1, including its amplitude and support when lifted.

B.1 Original zeta with both Mellin endpoints retained

At $t = 0$, HTB6 proves

$$K_0(x) = x^{-1/2} K_0(1/x) + \frac{1}{2} x^{-1/2} - \frac{1}{2}. \quad (\text{ZTC9})$$

Split ZTC4 at $x = 1$, apply this identity only to the integral over $(0, 1)$, and substitute $y = 1/x$ there. Initially $\operatorname{Re} s > 1$, and direct integration gives

$$\zeta(s) = \frac{\pi^{s/2}}{\Gamma(s/2)} \left\{ \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty K_0(x) \left(x^{s/2-1} + x^{-(s+1)/2} \right) dx \right\}. \quad (\text{ZTC10})$$

The two separate rational terms are the transforms of the two separate terms $x^{-1/2}/2$ and $-1/2$. The integral on $[1, \infty)$ is entire in s , since exponential decay bounds every logarithmic derivative on compact subsets of the s -plane. This proves the meromorphic continuation of the identity and keeps both endpoint contributions. Neither has been declared zero or absorbed into an unrecorded constant.

Proof of the Gaussian Poisson identity used above. For $x > 0$, periodize $e^{-\pi xy^2}$. All derivatives converge uniformly on a unit interval, by Gaussian bounds on the summands. The m th Fourier coefficient is its unfolded integral over $\mathbb{R}$, $x^{-1/2} e^{-\pi m^2/x}$. To verify that transform directly at scale one, differentiation under the integral and integration by parts give $\widehat{f}'(\eta) = -2\pi\eta\widehat{f}(\eta)$; the Gaussian integral gives $\widehat{f}(0) = 1$. Scaling gives the coefficient just stated. Their absolute convergence permits evaluation of the Fourier series at zero, giving $1+2K_0(x) = x^{-1/2}(1+2K_0(1/x))$. Rearranging with all terms retained gives ZTC9.